Experiment 6

Aim:

Apply K-Means and Hierarchical Clustering on sample datasets

Theory:

1. Dataset Source

Dataset Name: Mall Customer Segmentation Data
Source Link:
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

2. Dataset Description

The dataset contains information about customers visiting a mall. The goal is to segment customers into different groups based on their purchasing behavior using clustering techniques.

Dataset Characteristics

- **Total Instances:** 200
- **Number of Features:** 5
- **Target Variable:** None (Unsupervised Learning)
- **Task Type:** Clustering

Features Description

Feature	Description
CustomerID	Unique ID assigned to customer
Gender	Male / Female
Age	Age of the customer
Annual Income (k$)	Annual income in thousand dollars
Spending Score (1–100)	Score assigned based on spending behavior

Important Characteristics

- No missing values.

- Combination of categorical and numerical features.
- Commonly used features for clustering:
    - Annual Income
    - Spending Score

Since clustering is unsupervised learning, there is **no predefined target variable**.

3. Mathematical Formulation of the Algorithms

A. K-Means Clustering

K-Means partitions data into **K clusters** by minimizing intra-cluster variance.

Objective Function

$$J = \sum_{i=1}^{K} \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

Where:

K = Number of clusters
C_i = Cluster i
μ_i = Centroid of cluster i
x_j = Data points

Algorithm Steps

1. Choose K cluster centers randomly.
2. Assign each point to the nearest centroid using Euclidean distance:

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

3. Recalculate centroids.
4. Repeat until centroids stabilize.

B. Hierarchical Clustering

Hierarchical clustering builds a tree-like structure (dendrogram).

Types

- Agglomerative (Bottom-up)
- Divisive (Top-down)

In Agglomerative clustering:

- Start with each data point as a cluster.
- Merge closest clusters step by step.

Distance Between Clusters

Single Linkage:

$D(A,B)$=min($d(a,b)$)D(A,B)=min(d(a,b))

Complete Linkage:

$D(A,B)$=max($d(a,b)$)D(A,B)=max(d(a,b))

Average Linkage:

$D(A,B)$=1$|A||B|\sum d(a,b)$D(A,B)=|A||B|1∑d(a,b)

The final result is represented as a **dendrogram**.

4. Algorithm Limitations

K-Means Limitations

1. Must specify K beforehand.
2. Sensitive to initialization.
3. Works well only for spherical clusters.
4. Sensitive to outliers.
5. Not suitable for non-globular cluster shapes.

Hierarchical Clustering Limitations

1. Computationally expensive for large datasets.
2. Once clusters are merged, cannot be undone.
3. Sensitive to noise and outliers.
4. Difficult to determine optimal number of clusters.

5. Methodology / Workflow

Step 1: Data Collection

- Download dataset from Kaggle.

Step 2: Data Preprocessing

- Remove CustomerID.
- Encode Gender if used.
- Select relevant features (Annual Income & Spending Score).
- Perform feature scaling (StandardScaler).

Step 3: Apply K-Means

- Use Elbow Method to determine optimal K.
- Fit K-Means model.
- Assign cluster labels.

Step 4: Apply Hierarchical Clustering

- Plot dendrogram.
- Determine optimal clusters.
- Apply Agglomerative clustering.

Step 5: Visualization

- Plot clusters using scatter plots.
- Compare clustering results.

Workflow Diagram

Dataset Collection
↓
Data Cleaning & Feature Selection
↓
Feature Scaling
↓
K-Means Clustering
↓
Elbow Method
↓
Hierarchical Clustering
↓
Dendrogram Analysis
↓

Cluster Visualization & Comparison

6. Performance Analysis

Since clustering is unsupervised, evaluation is done using internal metrics.

Inertia (K-Means)

Measures within-cluster sum of squares.
Lower value indicates compact clusters.

Silhouette Score

$$S = \frac{b - a}{\max(a, b)}$$

Where:

- aa = Mean intra-cluster distance
- bb = Mean nearest-cluster distance

Range: -1 to +1

- Closer to +1 → Well-separated clusters
- Around 0 → Overlapping clusters
- Negative → Wrong clustering

Observations

- Elbow method typically suggests **K = 5** clusters.
- Silhouette score generally between **0.4 – 0.6**, indicating reasonable clustering.
- K-Means performs faster.
- Hierarchical clustering provides better visualization through dendrogram.

7. Hyperparameter Tuning

K-Means Parameters

- n_clusters (K)
- init (k-means++ or random)
- max_iter
- n_init

Tuning Method:

- Use Elbow method.
- Compare Silhouette scores for different K values.

Impact:

- Proper K improves separation.
- k-means++ improves centroid initialization.
- Increasing max_iter ensures convergence.

Hierarchical Clustering Parameters

- n_clusters
- linkage (single, complete, average, ward)

Tuning Method:

- Analyze dendrogram.
- Compare Silhouette score for different linkage methods.

Impact:

- Ward linkage often gives compact clusters.
- Complete linkage reduces chaining effect.

Code:

```
import os
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans, AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.preprocessing import LabelEncoder

plots_dir = 'exp6/plots'
os.makedirs(plots_dir, exist_ok=True)

# Load dataset
file_path = 'exp6/archive (5)/Mall_Customers.csv'
df = pd.read_csv(file_path)
```

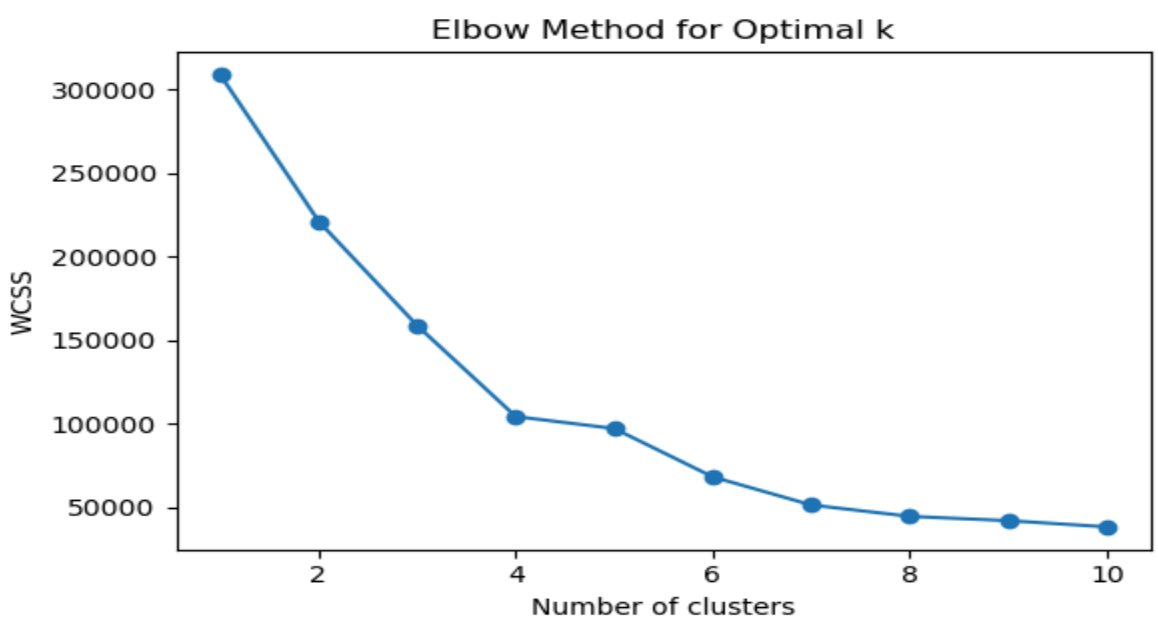
```
# Encode Gender
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])

# Select features for clustering
X = df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]

# --- K-Means Clustering ---
# Find optimal number of clusters using Elbow Method
wcss = []
for i in range(1, 11):
   kmeans = KMeans(n_clusters=i, random_state=42)
   kmeans.fit(X)
   wcss.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.tight_layout()
plt.savefig(os.path.join(plots_dir, 'elbow_method.png'))
plt.close()

# Fit KMeans with optimal clusters
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X)
df['KMeans_Cluster'] = y_kmeans

# Visualize KMeans clusters (using first two features for 2D plot)
plt.figure(figsize=(6,4))
for i in range(5):
   plt.scatter(X[y_kmeans == i]['Annual Income (k$)'], X[y_kmeans ==
i]['Spending Score (1-100)'], label=f'Cluster {i}')
plt.scatter(kmeans.cluster_centers_[:,1],
kmeans.cluster_centers_[:,2], s=200, c='black', marker='X',
label='Centroids')
plt.xlabel('Annual Income (k$)')
```

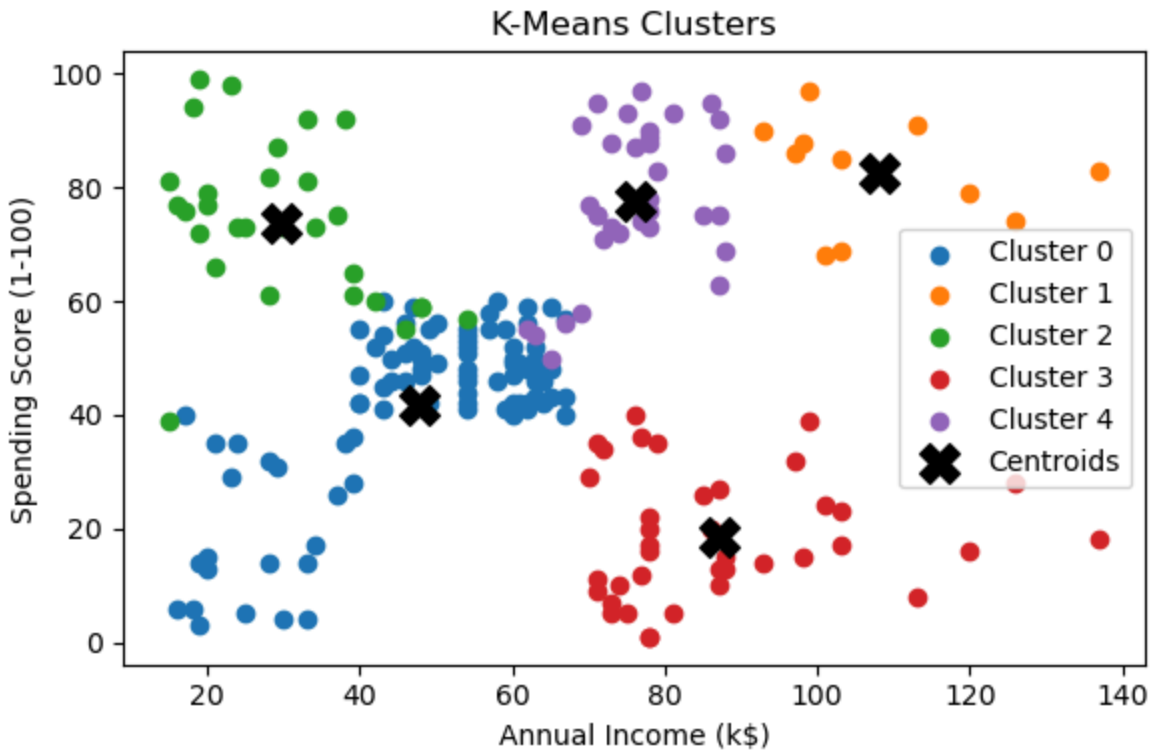
```
plt.ylabel('Spending Score (1-100)')
plt.title('K-Means Clusters')
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(plots_dir, 'kmeans_clusters.png'))
plt.close()

# --- Hierarchical Clustering ---
# Dendrogram to find optimal clusters
import numpy as np
sample_size = 50
if len(X) > sample_size:
   np.random.seed(42)
   sample_indices = np.random.choice(X.index, size=sample_size,
replace=False)
   X_sample = X.loc[sample_indices]
else:
   X_sample = X
plt.figure(figsize=(10, 5))
Z = linkage(X_sample, method='ward')
dendrogram(Z)
plt.title('Dendrogram (Sample of 50 Customers)')
plt.xlabel('Sampled Customers')
plt.ylabel('Euclidean distances')
plt.tight_layout()
plt.savefig(os.path.join(plots_dir, 'dendrogram_sample.png'))
plt.close()

# Fit Agglomerative Clustering
hc = AgglomerativeClustering(n_clusters=5, linkage='ward')
y_hc = hc.fit_predict(X)
df['Hierarchical_Cluster'] = y_hc

# Visualize Hierarchical clusters
plt.figure(figsize=(6,4))
for i in range(5):
   plt.scatter(X[y_hc == i]['Annual Income (k$)'], X[y_hc ==
i]['Spending Score (1-100)'], label=f'Cluster {i}')
```

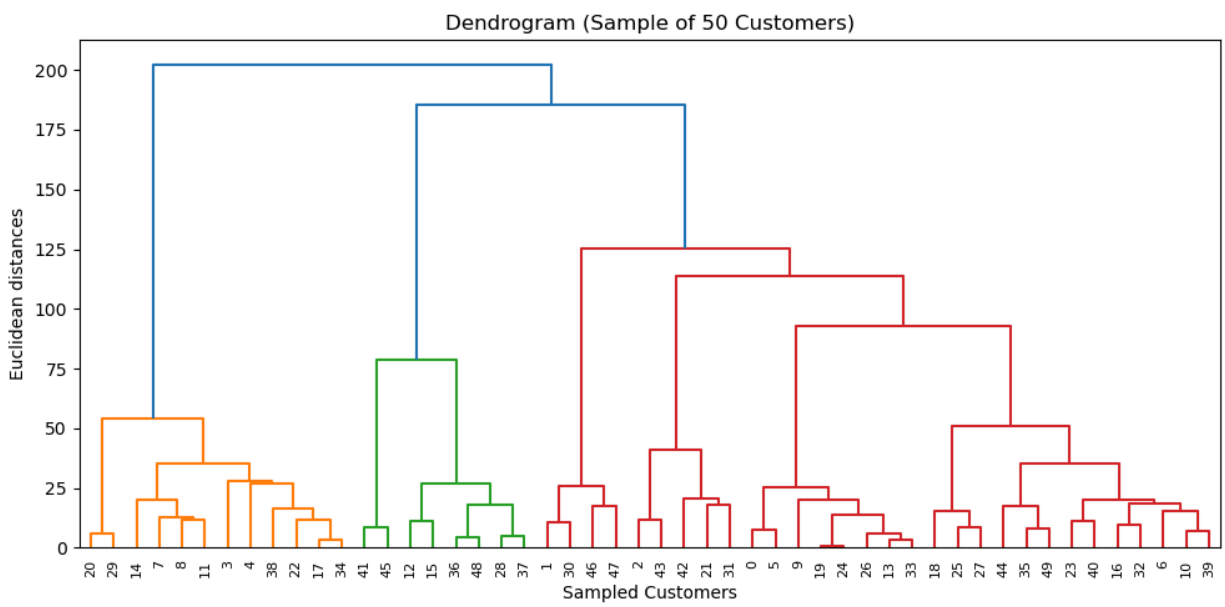
```
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Hierarchical Clusters')
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(plots_dir, 'hierarchical_clusters.png'))
plt.close()

# Display sample of results
print('Sample of clustered data:')
print(df.head())
```

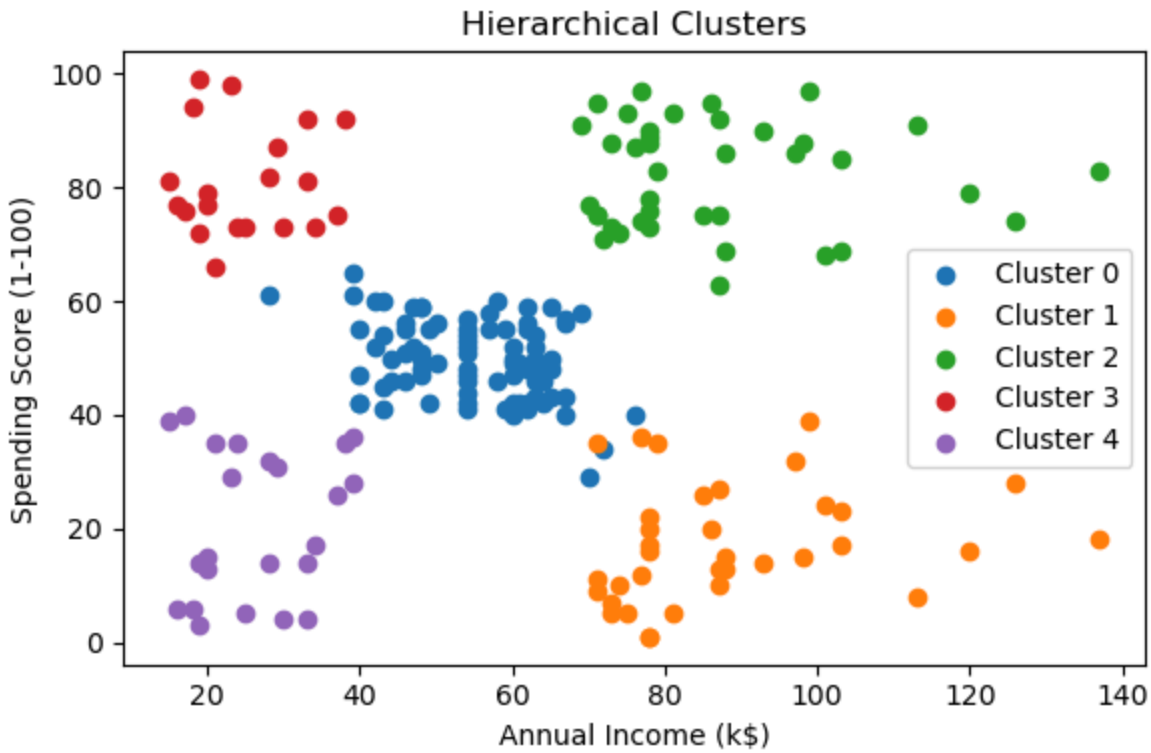
Output:

```
Sample of clustered data:
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  KMeans_Cluster  Hierarchical_Cluster
0           1       1   19                  15                      39               2                     4
1           2       1   21                  15                      81               2                     3
2           3       0   20                  16                       6               0                     4
3           4       0   23                  16                      77               2                     3
4           5       0   31                  17                      40               0                     4
```




K-Means Clusters
Spending Score (1-100)
Annual Income (k$)
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Centroids


Dendrogram (Sample of 50 Customers)
Euclidean distances
Sampled Customers



Conclusion:

In this experiment, K-Means and Hierarchical Clustering were applied to the Mall Customer Segmentation dataset to group customers based on income and spending behavior. The Elbow Method and Silhouette Score were used to determine the optimal number of clusters. K-Means provided faster computation and clear cluster separation, while Hierarchical Clustering offered better interpretability through dendrogram visualization.